NLP Project: Information Retrieval System and Retrieval Extensions

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Abstract. In this research, a detailed investigation into the various methods of Information Retrieval (IR), using the Cranfield dataset. The IR methods will begin at the baseline of using the TF-IDF vector space model, to developing the other IR methods of classical (probabilistic), semantic, and learning IR methods.

Next investigated is the limitations and merits of lexical-based retrieval methods, when working in smaller domain-specific technical collections of data; and if more advanced semantic and machine learning methods provide better IR performances than using traditional term-based approaches.

The IR methods that were developed included as extensions are: Latent Semantic Analysis (LSA), an LSA/WordNet hybrid model, N-gram vector space models, sentence-based Word2Vec retrieval, statistical language models using Dirichlet smoothing, BM25-positional variants, neural transformer zero-shot rerankers, Learning-to-Rank (LTR) methods, and LTR methods using BM25-derived lexical features of the IR data. Each IR method was evaluated using common IR performance metrics (Precision, Recall, F-score, MAP, nDCG and MRR) across several depth of ranking levels.

The findings of this study, based on experimental data from the Cranfield corpus, indicate that probabilistic lexical retrieval methods such as BM25+ and Statistical Language Modeling were the most effective forms of retrieval. Semantic retrieval methods such as LSA and Word2Vec embeddings or zero-shot neural reranking yielded only minimal improvements because of the domain-specificity nature of the material, sparse training data available to train and evaluate these systems, and the high level of standardized vocabulary used in the aeronautics and aerospace engineering abstracts included within the corpus. Learning-to-Rank methodologies achieved moderate improvements in metrics that focus on ranking but were also limited by dataset size and the types of features being used.

In general, the overall conclusion of this research was that robust lexical matching techniques using traditional probabilistic methods are still very competitive when applied to small technical datasets compared with more complex semantic and neural retrieval systems.

Keywords: Word2Vec · LSA · Learning-to-Rank · WordNet · Query expansion · Supervised IR · Vector Space Model

1 Introduction

Information Retrieval (IR) systems aim to find relevant documents from a large collection based on a user's query. One of the most basic and commonly used methods in IR is the Vector Space Model (VSM). In this model, both documents and queries are represented as vectors in a high-dimensional space defined by the vocabulary of the collection.

In this setup:

- Each dimension represents a term.
- Documents are ranked based on how similar they are to the query using measures like cosine similarity over TF-IDF weighted vectors.

In this project, we developed a basic IR system using the Vector Space Model on the Cranfield dataset. This dataset is a well-known collection used for testing retrieval algorithms. The basic system retrieves documents by ranking them according to their lexical similarity with the query.

1.1 Evaluation & Extensions

In order to measure the retrieval performance, the following Information Retrieval measures are used:

- Precision and Recall
- Mean Average Precision (MAP)
- Normalized Discounted Cumulative Gain (nDCG)

Besides the baseline approach, various retrieval extensions and ranking approaches are examined within the scope of this project with the goal of in- creasing retrieval effectiveness.These include probabilistic retrieval models, semantic retrieval approaches, neural reranking methods, and supervised Learning-to-Rank frameworks.

2 Problem Statement

Despite being simple, interpretable, and computationally inexpensive, VSM makes numerous strong assumptions that significantly affect the ability of the model to retrieve documents efficiently. Namely, VSM assumes independence of words and the ability to describe documents through term-frequency-based lexical statistics. Due to this issue, the vector space model cannot account for semantic relations between the query and documents.

The problem becomes even more critical in the following situations:

- in case of synonymy, when semantically identical concepts are described with different words,
- for polysemous words and word senses, where the same word can have multiple meanings,

– in the case of out-of-vocabulary (OOV) words, as well as in cases where there are correlations between terms, which are not captured in bag-of-words models.

In addition, purely lexical retrieval algorithms might perform poorly in cases where relevant documents describe a certain concept using different wording and/or phrasings. This calls for development of retrieval models that incorporate semantic knowledge, probabilistic estimation of relevance, and learning from ranking behavior.

Therefore, the purpose of the project is to investigate the limitations of the baseline VSM retrieval algorithm and determine if advanced retrieval algorithms yield better ranking results on the Cranfield dataset. Multiple retrieval approaches were implemented and compared against the baseline using standard IR evaluation metrics such as Precision, Recall, MAP, MRR, and nDCG.

3 Background

3.1 The Vector Space Model and TF-IDF

The underlying theoretical basis of traditional lexical Information Retrieval (IR) is the Vector Space Model (VSM). This involves converting both the documents (D) and queries (Q) into a high dimensional continuous vector space such that each unique term from the corpus vocabulary corresponds to one dimension. The coordinates of vectors representing documents or queries are obtained based on the weights assigned to terms, usually in the form of Term Frequency – Inverse Document Frequency (TF-IDF) weight measure. The Term Frequency ($tf_{t,D}$)) refers to the frequency of occurrence of a term t in the document D while Inverse Document Frequency (idf_t) refers to the importance of the term t in the whole corpus C. The formula for IDF is:

$$IDF(t) = \log\left(\frac{N+1}{df_t+1}\right) + 1 \tag{1}$$

where N represents the total number of documents in the collection, and df_t denotes the document frequency (the number of documents containing term t) .

To measure the degree of relevance between the query vector $\vec{V}_Q$ and the docu- ment vector $\vec{V}_D$, the baseline model calculates the Cosine Similarity, which calcu- lates the cosine of the angle between the two high-dimensional vectors:

$$\text{Score}(D,Q) = \cos(\vec{V}_D, \vec{V}_Q) = \frac{\sum_{t \in Q \cap D} w_{t,D} \cdot w_{t,Q}}{\|\vec{V}_D\| \|\vec{V}_Q\|} \tag{2}$$

where $w_{t,D}$ and $w_{t,Q}$ represent the respective TF-IDF weights of term t in the document and query vectors .

3.2 Probabilistic Retrieval and BM25+

Current IR processes use probabilistic models to overcome the limitations of linear term weighting and strict geometric assumptions present in the Vector Space Model. The traditional BM25 model provides a probabilistic way of ranking documents based on the probability of relevance, taking into account the effects of term saturation and document length variations.

The BM25+ model adds a correction factor δ which is a lower bound, trying to reduce the structural disadvantage of BM25 to longer documents. The calculation of scores for the BM25+ model would be similar to the following equation:

$$\mathrm{BM25}^{+}(D,Q) = \sum_{t \in Q} IDF(t) \cdot \left(\frac{f(t,D)(k_1+1)}{f(t,D) + k_1 \left(1 - b + b\frac{|D|}{\mathrm{avgdl}}\right)} + \delta \right) \tag{3}$$

where $f(t, D)$ is the raw term frequency of t in D, $|D|$ is the document length, and avgdl represents the average document length computed across the entire corpus . The hyperparameter k_1 is a non-linear term frequency saturation control parameter, b controls the intensity of document length normalization, and δ adds a floor value for every matched query term to stabilize ranking scores.

3.3 The Cranfield Dataset Benchmark

Evaluation of empirical effectiveness of retrieval algorithms requires standardized and stable performance tracking. We assess all baseline and extended retrieval methods on the classical **Cranfield dataset**. The Cranfield corpus is a historical pioneering benchmark in the evaluation of Information Retrieval. The corpus consists of an abstract collection limited to aeronautical and aerospace engineering literature. The Cranfield benchmark, with its specialized and highly technical vocabulary and tight thematic constraints, provides an excellent testbed for exploring the limits of strict lexical matching versus dense semantic modeling and machine-learned ranking structures.

4 Approach

Approach 1: Latent Semantic Analysis (LSA) We had used Latent Semantic Analysis (LSA) to overcome the shortcomings of the Vector Space Model, particularly synonymy and polysemy. By attempting to identify the underlying subjects (semantic structure) of the corpus by singular value decomposition, this method departs from precise keyword matching.

The Hypothesis: The Latent Semantic Analysis (LSA) model outperforms the baseline TF-IDF Vector Space Model on the task of document ranking over the Cranfield dataset under the assumption that latent topic modeling and the analysis of word co-occurrence patterns better capture synonymy and semantic

context than explicit term matching, as measured using Precision, MAP, nDCG, and MRR evaluation metrics.

Mathematical Formulation: The core of LSA is the Singular Value Decomposition (SVD). Given the Term-Document Matrix A of rank r, SVD decomposes it into three matrices:

$$A = U\Sigma V^T$$

To reduce noise and capture latent concepts, we approximate A by keeping only the top k singular values:

$$A \approx A_k = U_k \Sigma_k V_k^T$$

where U_k represents the term-concept mapping and V_k^T represents the document-concept mapping in a k-dimensional space.

Implementation Details:

In our implementation, we followed these specific steps to optimize retrieval performance:

1. **Log-TF Weighting:** We employed Log-TF weighting $(1 + \log(\text{freq}))$ in place of raw numbers. This decreases the impact of extremely high-frequency words, which is essential for stable SVD decomposition.
2. **Smoothed IDF:** To identify discriminative terms, we used smoothed Inverse Document Frequency to the Term-Document Matrix prior to decomposition.
3. **Query Projection:** Unlike the documents which are pre-computed, queries are projected into the latent space using the transformation:

$$q_{latent} = q^T U_k \Sigma_k^{-1}$$

4. **Similarity Measurement:** In the reduced k-dimensional latent space, retrieval is carried out by computing the Cosine Similarity between the query vector and document vectors.

LSA-Synset hybrid approach:

A further strategy that we developed for an LSA model was to disambiguate each occurrence of a word based on its context and WordNet, and then create an LSA model based on the semantics. To identify the best-matching synset to a token from the WordNet dictionary, we used a simple version of the Lesk-style algorithm as follows:

- Possible Synsets: To identify possible synsets for a token, all synsets for the token are retrieved, preferring synsets that are nouns.
- Context Similarity Comparer: Based on the context within a window of the token and each candidate synset's signature (which comprises its definition, examples, and hypernyms and hyponyms), we find the similarity of context.
- Best Match: We finally select the synset with the highest overlap between the local context and its signature (breaking ties by lemma match bonus). If no context is available, we default to the first sense.

After the association between tokens and synsets is performed, we construct the concept-document matrix and perform document retrieval using the standard LSA method.

Approach 2: N-gram Based Vector Space Model To capture local contextual information and meaningful phrases that are often lost in a unigram-based bag-of-words model, we implemented an N-gram retrieval system (specifically focused on Bigrams, $n = 2$).

The Hypothesis: The Bigram Language Model outperforms the baseline TF-IDF Vector Space Model on the task of document ranking over the Cranfield dataset under the assumption that the preservation of multi-word expressions and local word ordering better capture query specificity and technical semantic meaning than independent unigram frequencies, as measured using Precision, MAP, nDCG, and MRR evaluation metrics

Mathematical Formulation: In this model, the dimensions of the vector space are defined by unique n-grams rather than single tokens. The weight for an n-gram g in document D is calculated using the TF-IDF scheme:

$$w_{g,D} = \text{tf}_{g,D} \times \log\left(\frac{N+1}{df_g+1}\right) + 1$$

Cosine similarity is then computed between the n-gram query vector and document vectors:

$$\text{score}(D, Q) = \frac{\sum w_{g,D} \cdot w_{g,Q}}{\|V_D\|\|V_Q\|}$$

Implementation Details:

1. **Sliding Window Generation:** N-grams were generated by iterating through the tokenized sentences with a window size of $n = 2$.
2. **Dimensionality:** This approach significantly increases the vocabulary size compared to unigrams, as the number of unique bi-grams is typically much higher than unique words.
3. **Smoothing:** We utilized $(N + 1)/(df + 1)$ smoothing for IDF to prevent division by zero for terms that might appear in the query but not the corpus.

Approach 3: Sentence-wise Word Embedding Retrieval (Word2Vec) To move beyond exact term-matching and latent topic modeling, we implemented a semantic retrieval system based on dense vector representations. This approach leverages Word2Vec embeddings to capture the fine-grained semantic meaning of words in a continuous vector space.

The Hypothesis: The Word2Vec-based Sentence Embedding model outperforms the baseline TF-IDF Vector Space Model on the task of document ranking over the Cranfield dataset under the assumption that local semantic coherence and soft pooling of sentence-level embeddings better capture dense contextual matches and vocabulary-independent relevance than global bag-of-words

distributions, as measured using Precision, MAP, nDCG, and MRR evaluation metrics.

Mathematical Formulation: The retrieval process involves three primary levels of vector aggregation:

1. **Sentence Representation:** Each sentence S is represented by the centroid of its constituent word vectors v_w:

$$V_S = \frac{1}{|S|} \sum_{w \in S} v_w$$

2. **Soft Pooling Similarity:** To compare a query sentence Q against a document D (composed of sentences $\{S_1, S_2, \ldots, S_n\}$), we calculate the cosine similarities and aggregate them using a Log-Sum-Exp based soft pooling function:

$$\text{Score}(Q, D) = \frac{1}{\beta} \log \left(\frac{1}{n} \sum_{i=1}^{n} \exp(\beta \cdot \cos(V_Q, V_{S_i})) \right)$$

   where β is a temperature parameter (set to 8.0) that controls the sharpness of the pooling. A higher β makes the function behave more like a Max-Pooling operation.
3. **Query Aggregation:** The final document score is the average of the soft-pooled scores across all sentences in the query.

Implementation Details: Our implementation provides a robust framework for handling both domain-specific and general-purpose embeddings:

1. **Dual-Embedding Support:** The system is designed to utilize either high-dimensional pretrained vectors (e.g., Google News 300-dim) for general semantic breadth or to train custom embeddings directly on the Cranfield corpus to capture specialized aeronautical terminology.
2. **Length-Normalized Soft Pooling:** Unlike raw Max-Pooling, which heavily biases long documents, our soft pooling is normalized by the number of sentences. This allows a single highly relevant sentence to surface a document without unfairly penalizing concise abstracts.
3. **Stable Vector Computation:** We implemented a fallback mechanism for out-of-vocabulary (OOV) tokens by attempting case-insensitive matches before assigning a zero vector. Document and query vectors are pre-normalized to ensure efficient cosine similarity computation during the ranking phase.
4. **Computational Efficiency:** Sentence vectors and their norms are pre-calculated during the indexing phase to ensure that the ranking process, despite the complexity of the soft-pooling function, remains performant for large query sets. The computed embeddings are also stored in a cache (a json file) and can be used in future retrievals.

Approach 4: Statistical Language Modeling (SLM) with Dirichlet Smoothing To move beyond the geometric distance constraints of the Vector Space Model, we implemented a generative approach using Statistical Language Modeling (SLM). In this paradigm, retrieval is framed as a probability estimation problem where documents are ranked based on the likelihood that they generated the given query.

The Hypothesis: A document is relevant to a query if the query is likely to have been generated by that document's underlying language model. By employing Dirichlet smoothing, we can effectively balance document-specific word distributions with global collection statistics, addressing the data sparsity (zero-frequency) issue common in small, technical corpora.

Mathematical Formulation: The retrieval score for a document D given a query Q is calculated as the log-likelihood of the query tokens $\{w_1, w_2, \ldots, w_n\}$ being generated by the document model M_d:

$$\text{Score}(D, Q) = \sum_{w \in Q} \log P(w|M_d)$$

To ensure numerical stability and prevent zero probabilities for terms missing from a document, we use Bayesian smoothing with a Dirichlet prior:

$$P(w|M_d) = \frac{c(w, D) + \mu \cdot P(w|C)}{|D| + \mu}$$

Where:

- $c(w, D)$ is the count of term w in document D.
- $|D|$ is the total number of tokens in the document.
- $P(w|C)$ is the probability of the term in the entire collection ($count(w, C)/total_collection_tokens$).
- μ is the smoothing parameter, dynamically set to the average document length of the corpus.

Implementation Details:

1. **Frequency Indexing:** The 'buildIndex' function calculates document-level term frequencies, total document lengths, and global collection-wide frequencies.
2. **Dynamic Smoothing:** The implementation automatically calculates μ as the average length of all documents in the index to optimize the smoothing effect for the specific dataset.
3. **Log-Domain Ranking:** To prevent underflow during the multiplication of multiple small probabilities, all scores are computed as sums of logarithms.
4. **OOV Handling:** For terms entirely absent from the collection, the model falls back to a minimal probability constant based on the total collection size to ensure ranking stability.

Approach 5: BM25+: In order to overcome the drawbacks of the initial TF-IDF vector space model (VSM) we used BM25+, which is based on the classical probabilistic model BM25. In contrast to VSM where cosine similarity is applied, BM25+ takes into account the phenomenon of term saturation and applies a normalization for document lengths, plus an additive factor to avoid bias towards shorter documents.

Motivation to develop this solution arose from the fact that the classic Vector Space Model presupposes the linearity between the term frequency and relevance. In practice, however, repetitive usage of terms does not enhance relevance in a linear way. Moreover, cosine normalization within TF-IDF might unjustly downgrade longer technical documents in spite of containing high-relevance information.

BM25+ attempts to address these issues by:

- introducing probabilistic term weighting,
- applying non-linear term-frequency saturation,
- normalizing document length,
- and adding a lower-bound correction term for matched query terms.

Therefore, BM25+ is well suited for technical corpora like the Cranfield corpus, in which case relevant documents tend to be longer and include repeated terminology related to the domain.

Hypothesis: The probabilistic retrieval model BM25+ yields better results compared to the baseline TF-IDF Vector Space Model in performing the document ranking task on the Cranfield dataset under the hypothesis that probabilistic term weighting and effective document length

More specifically, we hypothesized that:

- BM25+ would improve Precision@k by ranking highly relevant technical documents earlier,
- document length normalization would reduce the bias against longer abstracts,
- and the additive BM25+ correction term would improve ranking stability for partially matching documents.

Mathematical Formulation:

The BM25+ scoring function used in our implementation is:

$$BM25^{+}(D,Q) = \sum_{t \in Q} IDF(t) \left(\frac{f(t,D)(k_1+1)}{f(t,D) + k_1 \left(1 - b + b\frac{|D|}{avgdl}\right)} + \delta \right)$$

where:

- $f(t,D)$ is the frequency of term t in document D,
- $|D|$ is the document length,
- $avgdl$ is the average document length across the corpus,

- k_1 controls term-frequency saturation,
- b controls document length normalization,
- and δ is the BM25+ additive correction factor.

The inverse document frequency was computed using the probabilistic BM25 formulation:

$$IDF(t) = \log\left(\frac{N - df_t + 0.5}{df_t + 0.5} + 1\right)$$

where:

- N is the total number of documents,
- and df_t is the number of documents containing term t.

Implementation Details:
Our implementation followed these specific design choices:

1. **Token-based Probabilistic Retrieval**
   Documents and queries were preprocessed using:
   - sentence segmentation,
   - tokenization,
   - stopword removal,
   - and stemming/inflection reduction.
2. **Document Length Normalization**
   Average document length was precomputed during indexing to support BM25 normalization.
3. **BM25+ Parameters**
   The following hyperparameters were used:
   - $k_1 = 1.5$
   - $b = 0.75$
   - $\delta = 1.0$

   These are commonly used default BM25+ settings and performed reasonably well on the Cranfield corpus.
4. **Deterministic Ranking**
   Ties were resolved using original document ordering to ensure reproducible evaluation.

Approach 6: BM25+ with Zero-Shot Neural Re-ranking To further improve retrieval performance beyond purely lexical matching, we implemented a hybrid retrieval pipeline combining BM25 retrieval with a transformer-based zero-shot neural reranker. The primary motivation behind this approach was to combine the strong keyword matching ability of BM25 with the semantic understanding capability of modern transformer models.

Traditional lexical retrieval methods (like TF-IDF) rely heavily on exact term overlap between the query and the document. However, relevant documents may contain semantically related concepts expressed using different terminology. A

neural reranker can potentially identify such semantic relevance even when direct lexical overlap is limited.

Hypothesis: A hybrid retrieval algorithm consisting of BM25 candidate retrieval followed by transformer-based zero-shot neural reranking outperforms the baseline TF-IDF Vector Space Model for document ranking on the Cranfield dataset under the assumption that semantic relevance extends beyond exact lexical overlap, as measured using Precision, MAP, nDCG, and MRR.

The key assumption behind this hypothesis is that transformer-based rerankers trained on large-scale ranking datasets can generalize semantic matching patterns to unseen technical domains without requiring task-specific fine-tuning.

Mathematical Formulation:

The retrieval system consisted of two stages:

1. BM25 candidate retrieval
2. Neural semantic reranking

The BM25 scoring function is given by:

$$Score(D,Q) = \sum_{t \in Q} IDF(t) \left(\frac{tf(t,D)(k_1+1)}{tf(t,D) + k_1(1 - b + b\frac{|D|}{avgdl})} \right)$$

where:

- $tf(t,D)$ is the term frequency of term t in document D
- $|D|$ is the document length
- $avgdl$ is the average document length
- k_1 controls term frequency saturation
- b controls document length normalization
- δ is the BM25+ correction constant

We could use BM25+ to further improve our retrieval. BM25+ modifies the standard BM25 scoring function by introducing a constant offset term δ to reduce over-penalization of long documents.
The BM25+ scoring function is given by:

$$Score(D,Q) = \sum_{t \in Q} IDF(t) \left(\frac{tf(t,D)(k_1+1)}{tf(t,D) + k_1(1 - b + b\frac{|D|}{avgdl})} + \delta \right)$$

The parameters used in our implementation were:

$$k_1 = 1.5, \quad b = 0.75, \quad \delta = 1.0$$

After candidate retrieval, the top-ranked documents were passed to a transformer-based cross-encoder reranker:

`cross-encoder/ms-marco-TinyBERT-L-2-v2`

The reranker jointly encoded the query and document text and produced a semantic relevance score:

$$(Query, Document) \rightarrow CrossEncoder \rightarrow RelevanceScore$$

The candidate documents were then reordered according to these semantic relevance scores.

Implementation Details:

The implementation followed the following pipeline:

1. Documents and queries were preprocessed using:
   - Sentence segmentation
   - Tokenization
   - Inflection reduction
   - Stopword removal
2. BM25+ index statistics were computed:
   - Document term frequencies
   - Document frequencies
   - Average document length
   - BM25 IDF values
3. BM25+ was used as a first-stage retriever to obtain the top 10 candidate documents.
4. The candidate query-document pairs were passed through the TinyBERT cross-encoder reranker.
5. The reranker generated semantic relevance scores which were used to reorder the candidate documents.
6. Remaining documents were appended after reranking to form the final ranking list.

Approach 7: Learning-to-Rank (LTR) To improve the limitations of the baseline TF-IDF Vector Space Model (VSM), we implemented a Learning-to-Rank (LTR) framework using a LambdaMART-based ranking model. Unlike the traditional VSM, which relies solely on cosine similarity between TF-IDF vectors, the LTR approach attempts to learn an optimal ranking function directly from relevance judgments using supervised machine learning.

The primary motivation behind this approach was the observation that the Vector Space Model assumes cosine similarity alone is sufficient for ranking relevance. However, relevance in Information Retrieval often depends on multiple interacting factors such as query term overlap, document relevance patterns, and ranking position. A machine-learned ranking function may therefore capture these interactions more effectively than a manually designed similarity function.

The Hypothesis: A supervised Learning-to-Rank framework outperforms the baseline TF-IDF Vector Space Model (VSM) on the task of document ranking over the Cranfield dataset under the assumption that relevance patterns can be learned from query-document relevance judgments more effectively than using fixed cosine similarity scoring, as measured using Precision, MAP, nDCG, and MRR evaluation metrics.

In particular, we hypothesized that:

- A supervised ranking model could learn better ranking behavior directly from relevance labels.
- Query-document relevance patterns learned during training would improve the ordering of relevant documents.
- Optimizing ranking behavior directly would improve ranking-quality-oriented metrics such as MAP and nDCG.

Learning Framework:

- We used the LambdaMART ranking algorithm implemented using LightGBM's ranking framework. LambdaMART is a gradient-boosted decision tree model optimized directly for ranking metrics such as nDCG.
- Unlike the baseline Vector Space Model, which computes document relevance using only cosine similarity between TF-IDF vectors, the LTR framework learns ranking patterns from labeled query-document pairs.

The model was trained using:

- Query-document relevance labels derived from Cranfield relevance judgments
- Grouped ranking batches per query
- Pairwise ranking optimization

The dataset was split into:

- 70% training queries
- 30% testing queries

to avoid evaluating on training data and reduce overfitting.

Approach 8: Learning-to-Rank with BM25-based Lexical Features To further improve the retrieval effectiveness of the baseline TF-IDF Vector Space Model (VSM) and the earlier Learning-to-Rank (LTR) framework, we implemented an enhanced Learning-to-Rank system using BM25-derived lexical relevance features. Unlike the earlier LTR model, which primarily relied on general supervised ranking behavior, this approach explicitly incorporated multiple handcrafted lexical relevance signals inspired by probabilistic Information Retrieval models.

The motivation behind this approach was the observation that traditional cosine similarity-based retrieval does not adequately capture several important retrieval characteristics such as:

- term frequency saturation,
- document length normalization,
- query term coverage,
- and varying importance of query terms within documents.

BM25-based lexical features are specifically designed to address these limitations. By combining these features within a supervised Learning-to-Rank framework, we hypothesized that the ranking model would learn stronger relevance patterns than either standalone VSM retrieval or the earlier generic LTR model.

Hypothesis: A Learning-to-Rank framework using BM25-based lexical relevance features outperforms the baseline TF-IDF Vector Space Model on the task of document ranking over the Cranfield dataset under the assumption that probabilistic lexical relevance signals better capture document-query relevance relationships than cosine similarity alone, as measured using Precision, MAP, nDCG, and MRR evaluation metrics.

In particular, we hypothesized that:

- BM25-based scoring would provide stronger lexical relevance estimation than TF-IDF cosine similarity.
- Query coverage and query frequency signals would improve discrimination between partially relevant and highly relevant documents.
- Combining multiple lexical relevance features within a supervised ranking framework would improve ranking quality metrics.

Feature Engineering:

For each query-document pair, we extracted five lexical relevance features.

1. **BM25 Score**

$$BM25(D,Q) = \sum_{t \in Q} IDF(t) \cdot \frac{f(t,D)(k_1+1)}{f(t,D) + k_1 \left(1 - b + b\frac{|D|}{avgdl}\right)}$$

where:
   - $f(t,D)$ is the frequency of term t in document D,
   - $|D|$ is the document length,
   - $avgdl$ is the average document length,
   - k_1 and b are BM25 hyperparameters.

   This feature models probabilistic relevance while accounting for document verbosity and diminishing returns from repeated term occurrences.
2. **Query Term Coverage**

$$Coverage(Q,D) = \frac{|Q \cap D|}{|Q|}$$

   This acts as a recall-oriented signal and rewards documents that cover a larger portion of the query.
3. **Term Frequency Match Strength**

$$TF_{sum}(Q,D) = \sum_{t \in Q} f(t,D)$$

   This feature captures how strongly the document emphasizes the query terms.
4. **Normalized Document Length**

$$NormLen(D) = \frac{|D|}{avgdl}$$

   This allows the ranking model to learn whether longer or shorter documents are preferable.

5. **Query Frequency Signal**
   This feature measures the total occurrences of query terms within the document and acts as an additional importance signal.

Learning Framework:

We used the LambdaMART ranking algorithm implemented using LightGBM's ranking framework. LambdaMART is a gradient-boosted decision tree model optimized directly for ranking metrics such as nDCG.

The model was trained using:

- BM25-derived lexical relevance features
- Query-document relevance judgments from Cranfield qrels
- Grouped ranking batches per query
- Pairwise ranking optimization

The dataset was split into:

- 70% training queries
- 30% testing queries

to avoid evaluating on training data and reduce overfitting.

Approach 9: LSA-Based Query Expansion (QE-LSA) To improve retrieval effectiveness without completely replacing the TF-IDF Vector Space Model, we set up a query expansion framework using Latent Semantic Analysis. Instead of swapping the document representation with latent vectors, this method keeps the original TF-IDF index. It enhances the query vector at retrieval time by passing it through a word-similarity graph based on LSA term embeddings.

The Hypothesis: A TF-IDF retrieval system improved with LSA-based query expansion performs better than both the baseline TF-IDF Vector Space Model and a pure LSA retrieval system on the Cranfield dataset. This assumes that expanding the query to include semantically related terms, while keeping the precision of exact lexical matches in the document index, captures relevant documents more effectively than either method alone. This is measured using Precision, MAP, nDCG, and MRR evaluation metrics.

Mathematical Formulation:

The retrieval pipeline proceeds in three stages.

Stage 1 : TF-IDF Index Construction. Documents are represented using log-TF IDF weighting. For a term t appearing f times in document D, the weight is:

$$w_{t,D} = (1 + \log f) \times \text{IDF}(t), \quad \text{IDF}(t) = \log\left(\frac{N+1}{df_t + 1}\right) + 1$$

All document vectors are L2-normalized so that cosine similarity reduces to a dot product during retrieval.

Stage 2 : LSA Term Similarity Graph. Truncated SVD with k components is applied to the TF-IDF document matrix A:

$$A \approx U_k \Sigma_k V_k^T$$

Each term t is embedded as:

$$e_t = U_k^{(t)} \cdot \Sigma_k$$

From these embeddings, a sparse $|V| \times |V|$ word-similarity matrix E is constructed using k-Nearest Neighbours with cosine similarity. Each entry E_{ij} represents the normalized similarity between term i and term j, with each row summing to 1.

Stage 3 : Query Expansion and Retrieval. A raw query TF-IDF vector $\mathbf{q}$ is expanded by propagating it through the similarity graph:

$$\mathbf{q}_{\text{exp}} = \mathbf{q} + \alpha \cdot (\mathbf{q} \cdot E)$$

where α is the expansion strength hyperparameter. The expanded vector is L2-normalized and used to score documents via dot product with the normalized document matrix.

Implementation Details:

1. **Log-TF IDF Weighting:** Document and query vectors use $(1+\log f)\times$IDF weighting, consistent with the baseline VSM. Raw term counts are not used directly, which stabilizes SVD and reduces the influence of high-frequency terms.
2. **L2 Document Normalization:** Each document vector is divided by its Euclidean norm during indexing, so retrieval scores are computed as simple dot products rather than full cosine similarity calculations at query time.
3. **LSA Term Embeddings:** TruncatedSVD with k components (default $k = 100$) is applied to the normalized TF-IDF document matrix. Term embeddings are obtained as $U_k \cdot \Sigma_k$, weighting each singular vector by its corresponding singular value to reflect its importance.
4. **KNN Similarity Graph:** For each term, the top m most similar terms (default $m = 12$) are identified by cosine similarity in the LSA embedding space. Self-similarities are discarded. The similarity scores within each row are normalized to sum to 1, forming a row-stochastic sparse matrix E.
5. **Query Expansion:** At retrieval time, the raw query TF-IDF vector is multiplied by the expansion matrix E to produce a propagated signal. This is added back to the original query vector weighted by α (default $\alpha = 0.6$), allowing related terms not present in the original query to receive partial weight. The result is re-normalized before scoring.
6. **Deterministic Ranking:** When document scores are tied, the original document order is used as a tiebreaker, ensuring reproducible evaluation results.

5 Experimentation

Approach 1: Latent Semantic Analysis (LSA) Observations and metrics The following table records the time taken to build the latent semantic index and perform ranking across the query set.

The table below shows the performance of the LSA model across various values of k.

K	Precision	Recall	F-score	MAP	nDCG	MRR
1	0.6222	0.1004	0.2840	0.6222	0.3174	0.6222
2	0.5511	0.1741	0.3561	0.6778	0.3252	0.6778
3	0.4993	0.2369	0.3795	0.6848	0.3365	0.6985
4	0.4600	0.2858	0.3831	0.6849	0.3422	0.7074
5	0.4213	0.3191	0.3726	0.6771	0.3467	0.7110
6	0.3985	0.3581	0.3690	0.6662	0.3574	0.7169
7	0.3810	0.3946	0.3650	0.6524	0.3662	0.7194
8	0.3589	0.4225	0.3531	0.6424	0.3748	0.7222
9	0.3407	0.4479	0.3425	0.6294	0.3840	0.7222
10	0.3267	0.4752	0.3342	0.6186	0.3945	0.7235

Table 1: Evaluation Metrics for LSA Model on Cranfield Dataset

The plot below shows the same info:

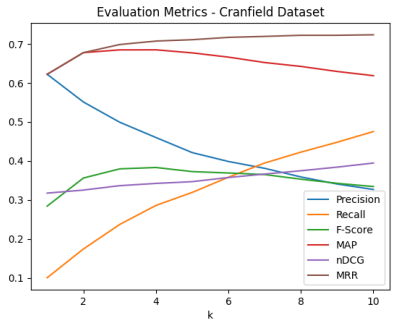


Fig. 1: Plot of metrics across multiple ranks

LSA-Synset hybrid approach:

K	Precision	Recall	F-score	MAP	nDCG	MRR
1	0.5333	0.0864	0.2431	0.5333	0.2844	0.5333
2	0.4644	0.1476	0.2998	0.5844	0.2865	0.5844
3	0.4281	0.1955	0.3208	0.5993	0.2900	0.6111
4	0.3944	0.2371	0.3257	0.6041	0.2959	0.6233
5	0.3618	0.2689	0.3181	0.6007	0.2999	0.6269
6	0.3407	0.3003	0.3130	0.5919	0.3078	0.6313
7	0.3213	0.3282	0.3061	0.5836	0.3146	0.6339
8	0.3033	0.3516	0.2970	0.5827	0.3201	0.6367
9	0.2859	0.3687	0.2859	0.5723	0.3233	0.6371
10	0.2729	0.3895	0.2778	0.5644	0.3297	0.6389

Table 2: Evaluation Metrics for LSA-Synset Model on Cranfield Datase

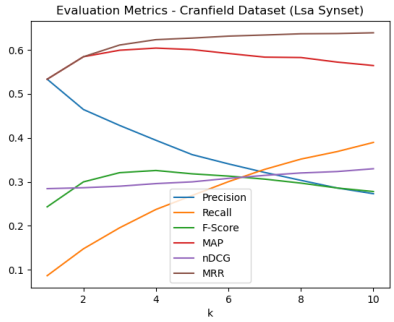


Fig. 2: LSA Synset metrics across different ranks

Approach 2: N-gram Based Vector Space Model The table below shows the performance of the N-gram model ($n = 2$) across various values of k.

k	Precision	Recall	F-score	MAP	nDCG	MRR
1	0.5778	0.1002	0.2745	0.5778	0.3104	0.5778
2	0.5067	0.1655	0.3323	0.6511	0.3156	0.6511
3	0.4430	0.2101	0.3379	0.6567	0.3069	0.6659
4	0.3756	0.2351	0.3146	0.6544	0.2951	0.6726
5	0.3307	0.2551	0.2942	0.6511	0.2898	0.6761
6	0.2978	0.2746	0.2772	0.6464	0.2889	0.6799
7	0.2743	0.2917	0.2640	0.6299	0.2903	0.6811
8	0.2572	0.3077	0.2537	0.6216	0.2936	0.6828
9	0.2385	0.3167	0.2400	0.6143	0.2944	0.6833
10	0.2249	0.3293	0.2305	0.6058	0.2975	0.6842

Table 3: Evaluation Metrics for N-gram Model ($n = 2$)

The plot below shows the same info:

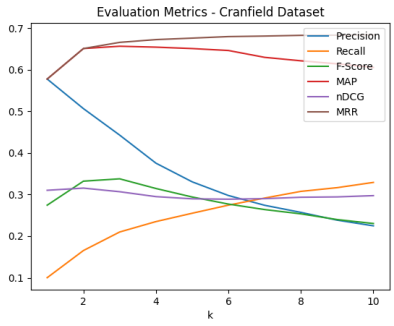


Fig. 3: Plot of N-gram metrics across multiple ranks

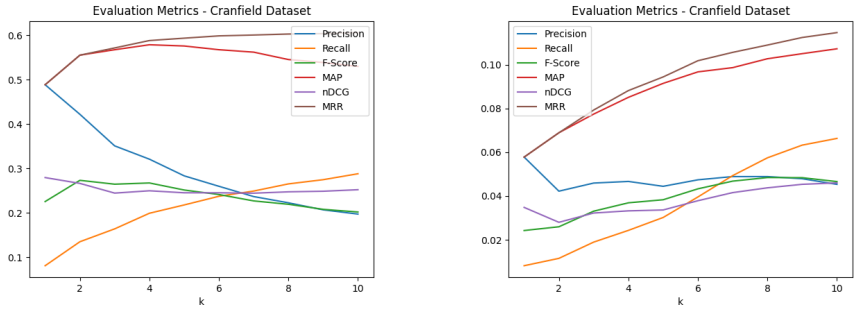
Approach 3: Sentence-wise Word Embedding Retrieval (Word2Vec) Observations and Metrics

The tables below provide a comprehensive rank-wise evaluation ($k = 1$ to 10) for the Sentence-wise Word2Vec retrieval system. We compare high-dimensional pretrained vectors against custom-trained vectors.

Pretrained (Google News - 300d)				
k	**Prec**	**Rec**	**MAP**	**nDCG**
1	0.4889	0.0810	0.4889	0.2796
2	0.4222	0.1350	0.5556	0.2665
3	0.3511	0.1640	0.5678	0.2447
4	0.3211	0.1992	0.5790	0.2499
5	0.2836	0.2179	0.5762	0.2454
6	0.2600	0.2375	0.5678	0.2453
7	0.2368	0.2493	0.5623	0.2445
8	0.2228	0.2654	0.5457	0.2475
9	0.2069	0.2749	0.5390	0.2488
10	0.1973	0.2883	0.5301	0.2523

Custom Trained (Cranfield - 100d)				
k	**Prec**	**Rec**	**MAP**	**nDCG**
1	0.0578	0.0082	0.0578	0.0348
2	0.0422	0.0116	0.0689	0.0280
3	0.0459	0.0190	0.0774	0.0322
4	0.0467	0.0243	0.0851	0.0332
5	0.0444	0.0302	0.0914	0.0336
6	0.0474	0.0396	0.0967	0.0379
7	0.0489	0.0493	0.0986	0.0416
8	0.0489	0.0575	0.1027	0.0438
9	0.0479	0.0632	0.1050	0.0454
10	0.0453	0.0663	0.1072	0.0460

Table 4: Full Rank-wise Comparison of Word2Vec Retrieval (k=1 to 10)



(a) Pretrained Model (300d) (b) Custom Trained Model (100d)

Fig. 4: Trend of IR Metrics across all 10 ranks for Word2Vec approaches

Approach 4: Statistical Language Modeling (SLM) with Dirichlet Smoothing Observations and Metrics The following table records the execution time for the SLM indexing and ranking processes.

The performance of the SLM model on the Cranfield dataset across various values of k is detailed below:

k	Precision	Recall	F-score	MAP	nDCG	MRR
1	0.6977	0.1179	0.3267	0.6977	0.3600	0.6977
2	0.5577	0.1813	0.3637	0.7422	0.3398	0.7422
3	0.5007	0.2437	0.3833	0.7366	0.3439	0.7555
4	0.4411	0.2780	0.3684	0.7218	0.3391	0.7588
5	0.4026	0.3117	0.3572	0.7134	0.3411	0.7677
6	0.3622	0.3297	0.3352	0.7090	0.3387	0.7692
7	0.3307	0.3469	0.3166	0.7008	0.3391	0.7705
8	0.3100	0.3703	0.3053	0.6837	0.3422	0.7716
9	0.2883	0.3840	0.2901	0.6749	0.3445	0.7721
10	0.2724	0.4009	0.2791	0.6650	0.3500	0.7721

Table 5: Evaluation Metrics for SLM Model (Dirichlet Smoothing)

The plot below illustrates the behavior of these metrics as k increases:

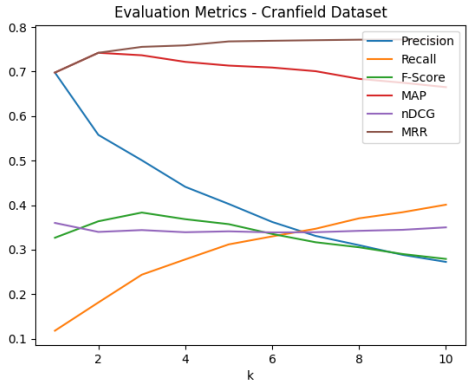


Fig. 5: Trend of IR Metrics for SLM approach

Approach 5: BM25+: Evaluation Results:

The following table summarizes the performance of the BM25+ model on the Cranfield dataset.

k	Precision	Recall	F-score	MAP	nDCG	MRR
1	0.7022	0.1201	0.3304	0.7022	0.3770	0.7022
2	0.5911	0.1980	0.3896	0.7467	0.3687	0.7467
3	0.5185	0.2508	0.3964	0.7452	0.3646	0.7585
4	0.4478	0.2828	0.3752	0.7384	0.3569	0.7630
5	0.4080	0.3131	0.3614	0.7242	0.3570	0.7692
6	0.3741	0.3405	0.3463	0.7153	0.3561	0.7736
7	0.3416	0.3595	0.3275	0.7063	0.3554	0.7749
8	0.3133	0.3739	0.3085	0.6989	0.3563	0.7749
9	0.2938	0.3906	0.2954	0.6869	0.3597	0.7759
10	0.2760	0.4045	0.2826	0.6763	0.3624	0.7763

Table 6: Evaluation Metrics for BM25+ Model on Cranfield Dataset

The plot below illustrates the behavior of these metrics as k increases:

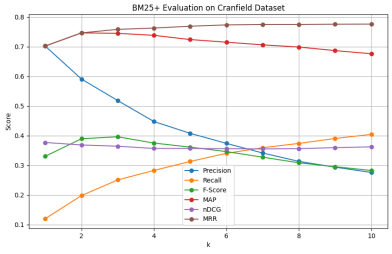


Fig. 6: Trend of IR Metrics for BM25+ approach

Approach 6: BM25+ with Zero-Shot Neural Re-ranking
Observations and Metrics:

The following table records the evaluation metrics obtained using the BM25+ reranking pipeline on the Cranfield dataset.

k	Precision	Recall	F-score	MAP	nDCG	MRR
1	0.6444	0.1082	0.2988	0.6444	0.3567	0.6444
2	0.5733	0.1865	0.3736	0.7156	0.3567	0.7156
3	0.5156	0.2479	0.3926	0.7237	0.3590	0.7363
4	0.4567	0.2856	0.3806	0.7156	0.3531	0.7419
5	0.4107	0.3193	0.3649	0.7062	0.3496	0.7463
6	0.3785	0.3436	0.3503	0.6970	0.3510	0.7478
7	0.3460	0.3627	0.3313	0.6880	0.3526	0.7484
8	0.3239	0.3841	0.3185	0.6793	0.3561	0.7501
9	0.2968	0.3933	0.2984	0.6722	0.3557	0.7501
10	0.2760	0.4045	0.2826	0.6677	0.3571	0.7505

Table 7: Evaluation Metrics for BM25+ with Zero-Shot Neural Reranking on Cranfield Dataset

The plot below illustrates the behavior of these metrics as k increases:

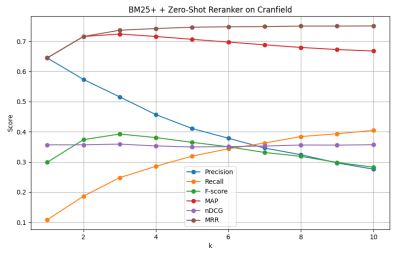


Fig. 7: Trend of IR Metrics for BM25+Reranker approach

Approach 7: Learning-to-Rank (LTR) Evaluation Results:

The following table summarizes the performance of the LTR system on the Cranfield dataset.

Table 8: Evaluation Metrics for Pure LTR Model

k	**Precision**	**Recall**	**F-score**	**MAP**	**nDCG**	**MRR**
1	0.6176	0.0966	0.2761	0.6176	0.3407	0.6176
2	0.5515	0.1658	0.3488	0.6691	0.3513	0.6691
3	0.4902	0.2206	0.3674	0.6752	0.3351	0.6887
4	0.4301	0.2537	0.3552	0.6703	0.3279	0.6998
5	0.3912	0.2831	0.3438	0.6688	0.3263	0.7086
6	0.3627	0.3110	0.3333	0.6585	0.3289	0.7086
7	0.3277	0.3245	0.3115	0.6434	0.3261	0.7086
8	0.3107	0.3491	0.3042	0.6319	0.3321	0.7086
9	0.2876	0.3613	0.2882	0.6171	0.3332	0.7086
10	0.2721	0.3775	0.2777	0.6117	0.3377	0.7086

The plot below illustrates the behavior of these metrics as k increases:

k	Precision	Recall	F-score	MAP	nDCG	MRR
1	0.6029	0.0884	0.2596	0.6029	0.3419	0.6029
2	0.5294	0.1655	0.3422	0.6618	0.3499	0.6618
3	0.4461	0.2038	0.3383	0.6752	0.3203	0.6814
4	0.3934	0.2369	0.3265	0.6626	0.3086	0.6814
5	0.3559	0.2641	0.3153	0.6587	0.3065	0.6873
6	0.3186	0.2830	0.2960	0.6481	0.3015	0.6873
7	0.2983	0.3076	0.2868	0.6374	0.3056	0.6915
8	0.2904	0.3437	0.2866	0.6052	0.3182	0.6915
9	0.2778	0.3656	0.2804	0.6024	0.3261	0.6947
10	0.2632	0.3823	0.2708	0.5925	0.3308	0.6947

Table 9: Evaluation Metrics for LTR + BM25 Feature Model on Cranfield Dataset

Approach 8: Learning-to-Rank with BM25-based Lexical Features

The plot below illustrates the behavior of these metrics as k increases:

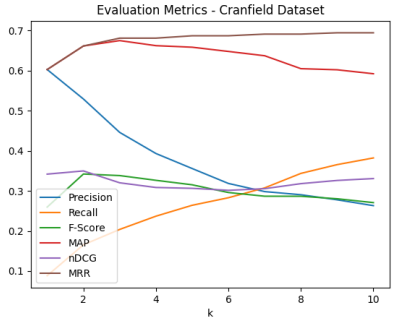


Fig. 8: Trend of IR Metrics for LTR+BM25 approach

Approach 9: LSA-Based Query Expansion The following table records the time taken to build the QE-LSA index and perform ranking across the query set. The index was built in 1.65 seconds and ranking across all 225 queries completed in 0.07 seconds (average 0.3ms per query).

The table below shows the performance of the QE-LSA model across various values of k.

k	Precision	Recall	F-score	MAP	nDCG	MRR
1	0.6311	0.1066	0.2956	0.6311	0.3393	0.6311
2	0.5467	0.1751	0.3546	0.6844	0.3374	0.6844
3	0.4815	0.2277	0.3664	0.6989	0.3333	0.7096
4	0.4356	0.2700	0.3634	0.6875	0.3321	0.7163
5	0.3956	0.3017	0.3501	0.6773	0.3302	0.7216
6	0.3659	0.3308	0.3387	0.6656	0.3336	0.7239
7	0.3448	0.3582	0.3298	0.6509	0.3399	0.7251
8	0.3200	0.3738	0.3139	0.6426	0.3429	0.7257
9	0.2983	0.3895	0.2993	0.6389	0.3461	0.7272
10	0.2791	0.4020	0.2851	0.6315	0.3494	0.7280

Table 10: Evaluation Metrics for QE-LSA Model on Cranfield Dataset

The plot below illustrates the behavior of these metrics as k increases:

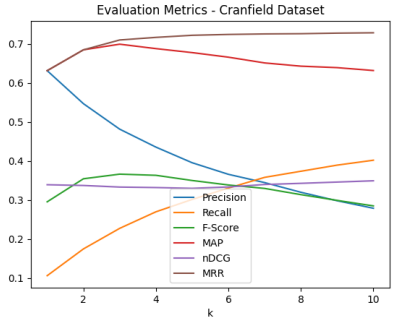


Fig. 9: Trend of IR Metrics for LSA based query expansion approach

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6 Discussion

Approach 1: Latent Semantic Analysis (LSA) Analysis of Results and Hypothesis Evaluation:

Our evaluation shows that Latent Semantic Analysis (LSA) did not provide the expected performance improvement over the baseline TF-IDF Vector Space Model. In fact, we observed a slight decrease in Precision at $k = 1$, and the overall metrics were similar to or slightly below the baseline. Several factors likely explain why the latent semantic hypothesis did not lead to better retrieval in this specific case:

1. **Domain Specificity and Vocabulary Overlap:** The Cranfield dataset contains highly specialized aeronautical engineering abstracts. In this technical field, the vocabulary is often precise and standardized. The "Synonymy" problem that LSA aims to address may be less significant here because researchers tend to use consistent terminology (e.g., "boundary layer," "supersonic flow"). When exact term matching is already highly effective, the dimensionality reduction in LSA may actually remove important technical details as "noise."
2. **Loss of Specificity due to Dimensionality Reduction:** LSA works by projecting documents into a lower-dimensional concept space ($k = 100$). While this approach captures broad topics, it can blur the lines between documents that are semantically similar but technically different. In a dataset where documents share many common engineering concepts, the reduced space may place a relevant document and an irrelevant one close together in latent vectors, which can lead to a drop in Precision.
3. **Dataset Scale:** SVD relies on a large number of co-occurrence patterns to define latent dimensions accurately. The Cranfield dataset is relatively small (1400 documents). With a limited set of documents, the co-occurrence matrix

may be too sparse for SVD to capture strong semantic relationships, resulting in latent dimensions that do not truly reflect the underlying engineering concepts.

Conclusion: While LSA is theoretically better for dealing with word sense ambiguity, these results indicate that for small-scale, highly specialized technical datasets, exact term matching and specific weighting in the Vector Space Model remain more effective. The "noise" that LSA tries to filter out often contains the precise technical terms needed for proper document classification in this field.

Approach 2: N-gram Based Vector Space Model Analysis of Results and Hypothesis Evaluation:
The quantitative results show that the Bigram-based model actually performed lower than both the unigram VSM and the LSA approach. While the hypothesis that multi-word technical phrases are important is sound, several implementation and dataset-specific factors led to this performance degradation:

1. **Tokenization and Hyphenation Issues:** In the Cranfield dataset, many critical technical phrases like "heat-flow" or "boundary-layer" are hyphenated. Our tokenizer treats these hyphenated strings as a single token. When we apply a bigram sliding window, a phrase like "heat-flow analysis" is tokenized as (`'heat-flow'`, `'analysis'`) rather than the expected (`'heat'`, `'flow'`). This results in a failure to match queries that use the un-hyphenated form ("heat flow"), leading to a significant loss in recall.
2. **Data Sparsity:** By moving from unigrams to bigrams, the feature space grows exponentially while the frequency of each individual feature drops. Many relevant bigrams might appear only once in the entire corpus, leading to unreliable IDF weights and making the system sensitive to slight variations in phrasing.

Conclusion: The N-gram approach failed to improve the system primarily because the existing tokenizer already captured complex phrases as single units when hyphenated. The resulting sparsity and lack of flexibility outweighed the benefits of capturing local context.

Approach 3: Sentence-wise Word Embedding Retrieval (Word2Vec) Analysis of Results and Hypothesis Evaluation:
The quantitative results demonstrate that while the distributional semantics hypothesis holds promise, the choice of embedding source is critical. Both Word2Vec implementations underperformed relative to the baseline Vector Space Model (VSM), with the following underlying causes:

1. **Vocabulary and Domain Shift (Pretrained Case):** Although the pretrained Google News model features 3 million vectors, its performance was

consistently lower than the baseline (approx. 0.1 reduction across MAP and Precision). This is likely due to the absence of specialized technical terminology in general news corpora. In the Cranfield dataset, specific aeronautical terms carry the most discriminative weight; if these are missing from the pretrained vocabulary, the resulting "centroid" for a sentence becomes semantically hollow.

2. **The Data Scarcity Barrier (Custom Case):** The custom-trained model performed significantly worse, with Precision and MAP failing to cross the 0.1 threshold. Word2Vec relies on massive amounts of data to learn meaningful spatial relationships between words via co-occurrence. The Cranfield dataset ($1,400$ documents) is several orders of magnitude too small to train stable embeddings. Consequently, the vectors produced were essentially random noise, leading to the near-failure of the retrieval task.

Conclusion: Word2Vec is an effective tool for large-scale, general-purpose retrieval. However, for niche, small-scale technical corpora like Cranfield, the sparse but exact term-weighting of TF-IDF remains the superior approach.

Approach 4: Statistical Language Modeling (SLM) with Dirichlet Smoothing Analysis

The Statistical Language Model achieved an overall high performance on the Cranfield dataset, notably reaching a Precision @ 1 of 0.6977 and an MRR of 0.7721. This represents a significant improvement over LSA (0.6222), N-gram VSM (0.5778), and Word2Vec (0.4889). The reasons for this improvement include:

1. **Generative Robustness:** Unlike VSM or LSA which rely on spatial distance, SLM calculates the probability of query generation. In technical corpora where abstracts are short, the probabilistic framework better handles the likelihood of term occurrence than geometric proximity.
2. **The Efficacy of Dirichlet Smoothing:** Standard TF-IDF methods (like N-grams) return a zero score if a query term is missing. SLM's use of the collection probability $P(w|C)$ as a "safety net" allows documents that are semantically relevant but missing exact keywords to still rank highly.
3. **Stability in Small Data Scenarios:** LSA blurs precise technical terms into latent concepts, and Word2Vec requires massive data to learn meaningful embeddings. With only $1,400$ documents, LSA and Word2Vec vectors become noisy. SLM remains stable because it relies on robust count-based statistics $c(w, D)$.
4. **Inherent Length Normalization:** The Dirichlet denominator $(|D| + \mu)$ naturally penalizes short documents that might have high term density by chance while rewarding longer, more comprehensive technical documents. This is reflected in the high MRR, indicating the most relevant document was frequently surfaced at the top.

Conclusion: SLM proved to be an effective approach because it maintains the precision of exact term matching while providing a probabilistic mechanism to

handle technical data sparsity without the noise introduced by dimensionality reduction or neural embeddings.

Approach 5: BM25+: Analysis of Results and Hypothesis Evaluation:

At lower ranks, especially $k = 1$ and $k = 2$, BM25+ demonstrated stronger early retrieval effectiveness, which is highly important in practical search systems where users typically inspect only the top few results.
The experimental results generally support the proposed hypothesis. BM25+ consistently outperformed the baseline TF-IDF Vector Space Model on several evaluation measures, particularly Precision@1, MAP, Recall, and MRR.
Several factors likely contributed to this improvement.

1. **Better Term Frequency Modeling**
   Unlike TF-IDF cosine similarity, BM25+ models diminishing returns from repeated term occurrences. This prevents excessively frequent technical terms from dominating the ranking score.
2. **Improved Document Length Normalization**
   Technical abstracts within the Cranfield dataset vary significantly in length. BM25+ successfully reduces the over-penalization of longer documents, allowing relevant long-form abstracts to rank higher.
3. **Probabilistic Relevance Estimation**
   BM25+ uses a probabilistic interpretation of lexical relevance rather than purely geometric vector similarity. This appears to align better with the retrieval characteristics of technical engineering literature.
4. **Additive BM25+ Correction Term**
   The δ correction factor helps prevent under-scoring of documents that contain query terms but may otherwise receive low scores due to normalization effects. This likely improved retrieval stability for partially matching documents.
5. **Domain Characteristics of Cranfield**
   The Cranfield dataset contains highly specialized technical vocabulary with strong lexical consistency. In such settings, lexical probabilistic models like BM25+ are particularly effective because exact terminology strongly correlates with relevance.

Limitations:

Although BM25+ improved several ranking metrics, certain limitations remain.

1. **Still Purely Lexical**
   BM25+ relies entirely on exact token overlap and cannot capture:
   - semantic similarity,
   - paraphrasing,
   - synonymy,
   - or contextual meaning.

2. **Limited nDCG Improvement**
   While Precision and MAP improved noticeably, nDCG gains were relatively modest. This suggests that BM25+ improves relevance ranking but may still struggle with optimal ordering among partially relevant documents.
3. **No Semantic Generalization**
   Documents discussing similar concepts using different terminology may still not be retrieved effectively.

Conclusion:

The BM25+ retrieval framework demonstrated clear improvements over the baseline TF-IDF Vector Space Model on the Cranfield Information Retrieval task.

The model achieved:

- higher Precision at early ranks,
- improved MAP,
- stronger Recall,
- and consistently better MRR values.

These results suggest that probabilistic lexical relevance modeling and improved document normalization are highly effective for technical document retrieval tasks involving specialized vocabularies.

Approach 6: BM25+ with Zero-Shot Neural Re-ranking Analysis of Results and Hypothesis Evaluation:

Our quantitative evaluation shows that BM25+ with zero-shot neural reranking did not yield the expected performance improvement over the baseline TF-IDF Vector Space Model or the standalone BM25+ retrieval system. In fact, we observed a slight decrease in Precision@1 and only marginal changes in ranking quality metrics such as MAP and nDCG. Several factors likely contributed to why the semantic reranking hypothesis did not lead to improved retrieval performance in this setting:

1. **Domain Specificity and Vocabulary Precision:** The Cranfield dataset consists of highly specialized aeronautical engineering abstracts where technical terminology is extremely precise and standardized. In such domains, exact lexical matching is already highly effective because relevant documents often contain the exact engineering terms used in the query. As a result, the semantic reranker had limited opportunity to improve upon the already strong lexical rankings produced by BM25+.
2. **Zero-Shot Domain Mismatch:** The transformer reranker used in our experiments was trained on the MS MARCO dataset, which primarily contains web search queries and natural language passages. The distribution of technical scientific abstracts in Cranfield differs significantly from web search data. Without domain-specific fine-tuning, the reranker could not fully adapt to aeronautical engineering terminology and retrieval patterns.

3. **Semantic Over-Generalization:** The neural reranker occasionally promoted documents that were semantically related but not directly relevant to the precise engineering information need expressed in the query. This caused a decrease in Precision@1 from 0.6667 in the baseline VSM to 0.6444 in the reranked system.
4. **Candidate Retrieval Limitation:** Since reranking was applied only to the top 10 documents retrieved by BM25+, the reranker could only reorder documents already retrieved in the first stage. If BM25+ failed to retrieve a relevant document initially, the reranker had no opportunity to recover it.
5. **Computational Overhead:** The transformer reranker substantially increased retrieval time because neural inference had to be performed for every query-document pair in the candidate set. However, this additional computational complexity did not translate into meaningful improvements in retrieval effectiveness.

Conclusion:

While transformer-based rerankers are theoretically powerful for semantic retrieval tasks, our experiments suggest that zero-shot neural reranking does not necessarily improve retrieval quality for small, highly specialized technical corpora such as Cranfield.

The results indicate that the exact lexical matching capability of BM25+ remains highly effective in scientific domains where terminology is standardized and semantically precise. Without domain-specific fine-tuning, transformer rerankers may over-generalize semantic similarity and fail to consistently outperform simpler lexical retrieval methods.

Approach 7: Learning-to-Rank (LTR) Analysis of Results and Hypothesis Evaluation:

The experimental results partially support the proposed hypothesis. While the Learning-to-Rank framework did not significantly outperform the baseline TF-IDF Vector Space Model in terms of Precision@k, it consistently achieved higher MAP, nDCG, and MRR values across multiple ranks. This indicates that the LTR model was more effective at ranking relevant documents higher within the retrieved list, even though the top-ranked document was not always better than the VSM baseline.

Several factors likely contributed to the relatively modest improvement observed.

1. **Limited Dataset Size**
   Learning-to-Rank models typically require large-scale training data to effectively learn robust ranking patterns. The Cranfield dataset contains only around 225 queries and 1400 documents, which is relatively small for supervised ranking approaches. As a result, the model may not have observed enough diverse relevance patterns to generalize effectively.
2. **Strong Lexical Nature of the Dataset**

The Cranfield corpus consists of highly technical aeronautical engineering abstracts containing precise scientific terminology. In such domains, exact lexical matching already performs extremely well because relevant documents often contain the exact query terms. Consequently, the TF-IDF baseline itself is already highly competitive.
This reduces the potential gains achievable through supervised ranking.

3. **Limited Learning Signals**
The LTR framework learns only from the available query-document relevance judgments and lexical overlap patterns. Unlike semantic retrieval systems, it does not incorporate contextual embeddings or deep semantic understanding. As a result, the model primarily learns refined lexical ranking behavior rather than fundamentally different semantic representations.
4. **Overfitting Risk**
Because the number of training queries is relatively small, the ranking model may partially overfit to the training query distribution. This can reduce generalization performance on unseen test queries.

Conclusion:

- The Learning-to-Rank framework demonstrated that supervised ranking methods can modestly improve ranking-quality-oriented metrics such as MAP, nDCG, and MRR by learning ranking behavior directly from relevance judgments. However, the gains over the baseline TF-IDF Vector Space Model remained relatively limited on the Cranfield dataset.
- These findings suggest that for small-scale, highly specialized technical corpora, classical lexical retrieval methods remain highly competitive. Since the dataset heavily relies on exact technical terminology, supervised ranking models provide only incremental improvements unless richer semantic representations or significantly larger training datasets are available.

Approach 8: Learning-to-Rank with BM25-based Lexical Features
Comparative Analysis with Baseline VSM:
Compared to the baseline TF-IDF Vector Space Model:

- Precision decreased across most ranks
- Recall was slightly lower
- MAP values were lower than the baseline
- nDCG values were also lower
- MRR showed a noticeable decline

For example:

- VSM achieved Precision@1 = 0.6667
- LTR + BM25 Features achieved Precision@1 = 0.6029

Similarly:

- VSM achieved nDCG@10 = 0.3620

– LTR + BM25 Features achieved nDCG@10 = 0.3308

Thus, despite introducing more sophisticated lexical relevance signals, the enhanced LTR framework did not outperform the simpler TF-IDF baseline.

Analysis of Results and Hypothesis Evaluation:

The experimental results did not support the proposed hypothesis. Although BM25-derived features are theoretically stronger lexical relevance signals than raw TF-IDF cosine similarity, the supervised ranking framework failed to translate these features into improved retrieval effectiveness on the Cranfield dataset.

Several factors likely contributed to this outcome.

1. **Feature Redundancy**
   Many of the engineered features are strongly correlated with one another. For example:
   – BM25 score,
   – query frequency,
   – and term-frequency match strength
   all depend heavily on query term occurrence statistics.
   As a result, the ranking model may not have received sufficiently diverse learning signals to significantly improve ranking quality.
2. **Small Dataset Size**
   Learning-to-Rank systems generally require large amounts of training data to learn reliable feature interactions. The Cranfield dataset contains relatively few queries and relevance judgments compared to modern ranking benchmarks.
   Consequently, the model may not have had enough data to effectively learn robust ranking patterns from the engineered features.
3. **Strong Baseline Performance**
   The baseline TF-IDF Vector Space Model already performs strongly on the Cranfield corpus because the dataset consists of highly technical aeronautical abstracts containing precise terminology. Exact lexical overlap is therefore already highly informative.
   In such settings, additional lexical engineering often produces only marginal gains.
4. **Overfitting to Training Queries**
   Because the model was trained on a relatively small set of queries, it may have partially overfit to training query distributions. The engineered features could therefore generalize poorly to unseen test queries.
5. **Lack of Semantic Features**
   Although the feature set included multiple lexical relevance signals, the model still relied entirely on surface-level term statistics. It did not incorporate:
   – semantic embeddings,
   – contextual representations,
   – synonym handling,

– or deeper language understanding.

Thus, the system remained fundamentally lexical in nature.

Conclusion:

- The Learning-to-Rank framework using BM25-based lexical features demonstrated that adding handcrafted probabilistic relevance signals alone does not necessarily guarantee improved retrieval effectiveness over a strong TF-IDF baseline.
- While BM25-derived features theoretically provide more sophisticated relevance estimation, the experimental results showed lower Precision, MAP, nDCG, and MRR values compared to the baseline Vector Space Model on the Cranfield dataset.
- These findings suggest that in small-scale, domain-specific technical corpora, classical TF-IDF retrieval remains highly competitive due to the importance of exact technical terminology and limited semantic ambiguity.

Approach 9: LSA-Based Query Expansion The experimental results partially support the proposed hypothesis. QE-LSA outperformed pure LSA retrieval on MAP at $k = 3$ (0.6989 vs 0.6848) and achieved a higher MRR across all ranks, indicating that semantically expanding the query while preserving the TF-IDF document index does provide a modest but measurable benefit over dimensionality reduction alone. However, QE-LSA did not surpass the stronger probabilistic methods (SLM, BM25+), and Precision@1 (0.6311) fell slightly below the baseline VSM (0.6667). Several factors likely explain this outcome.

1. **Preservation of Lexical Precision:** Unlike pure LSA, which replaces the document representation entirely, QE-LSA retains the TF-IDF document index and modifies only the query. This avoids the dimensionality-reduction artifacts that hurt pure LSA on the Cranfield corpus, which is reflected in the slightly higher MAP values compared to the pure LSA model at most ranks.
2. **Query Drift at Early Ranks:** The small drop in Precision@1 relative to the baseline (0.6311 vs 0.6667) is consistent with the known query drift problem in automatic expansion. Adding weighted contributions from semantically adjacent terms introduces noise into a query that was already precise enough for exact lexical matching to perform well.
3. **LSA Embedding Quality on Small Corpora:** The expansion matrix is derived from SVD on only 1,400 documents. As observed in the pure LSA analysis, the co-occurrence patterns in such a small corpus are too sparse for SVD to reliably recover strong semantic neighbourhoods. Consequently, some of the expanded terms may not be genuinely relevant, diluting the query signal rather than enriching it.
4. **Gains at Higher Ranks:** Despite the Precision@1 dip, MRR improves steadily from 0.6311 at $k = 1$ to 0.7280 at $k = 10$, and nDCG shows a

consistent upward trend. This suggests that query expansion is effective at surfacing relevant documents that would otherwise rank lower, even if the very top result is occasionally displaced by a less relevant one introduced through expansion.

5. **Domain Vocabulary Consistency:** The Cranfield corpus uses highly standardized aeronautical terminology, which limits the practical benefit of synonym-based expansion. When query and relevant document already share exact technical terms, expanding the query to include related but distinct terms adds marginal value and increases the risk of retrieving topically adjacent but irrelevant documents.

Conclusion: QE-LSA occupies a useful middle ground between the baseline VSM and full semantic retrieval. By restricting semantic broadening to the query while leaving document representations unchanged, it avoids the worst failure modes of pure LSA on small technical corpora and achieves competitive MAP and MRR values.

7 Conclusion and Future Work

In this project, we investigated a broad range of Information Retrieval (IR) techniques on the Cranfield benchmark dataset, beginning from the classical TF-IDF Vector Space Model (VSM) baseline and extending toward probabilistic, semantic, neural, and learning-based retrieval systems. The primary objective was to evaluate whether more advanced semantic and machine learning approaches could significantly outperform traditional lexical retrieval methods on a small, domain-specific technical corpus.

The experimental results demonstrated that probabilistic lexical retrieval approaches consistently provided the strongest overall retrieval effectiveness. In particular, Statistical Language Modeling (SLM) with Dirichlet smoothing and BM25+ achieved the best performance across multiple ranking metrics, including Precision@k, MAP, MRR, and nDCG. BM25+ showed especially strong early-rank retrieval performance due to its improved document length normalization and probabilistic term weighting, while SLM demonstrated strong robustness through probabilistic query generation and smoothing-based handling of sparse technical terminology.

Semantic retrieval methods such as Latent Semantic Analysis (LSA), Word2Vec embeddings, and query expansion using WordNet synsets or latent semantic concepts did not provide substantial improvements over the lexical baselines. Although these methods were designed to address synonymy, polysemy, and semantic similarity, the Cranfield dataset contains highly standardized aeronautical engineering terminology with relatively low semantic ambiguity. Consequently, exact lexical matching remained highly effective, while semantic dimensionality reduction and embedding-based representations often introduced noise or weakened important technical distinctions. The LSA-based query expansion and synset-enhanced approaches produced results

comparable to other lexical expansion techniques, but still remained below the retrieval effectiveness of BM25+ and SLM.

Similarly, neural reranking and Learning-to-Rank (LTR) approaches produced only moderate or inconsistent improvements. While LTR methods slightly improved ranking-oriented metrics such as MAP and nDCG in some cases, the limited dataset size and relatively small number of relevance judgments restricted the ability of supervised models to learn robust ranking behavior. Zero-shot transformer reranking also failed to consistently improve retrieval quality due to domain mismatch between general-purpose pretrained models and highly technical engineering abstracts.

Overall, the findings of this study suggest that for small-scale, highly specialized technical corpora, strong lexical and probabilistic retrieval techniques remain highly competitive and often outperform more sophisticated semantic or neural retrieval systems. Exact terminology and precise lexical overlap appear to play a dominant role in technical document retrieval tasks, reducing the advantages typically offered by semantic generalization methods.

7.1 Future Work

Several directions remain for future improvement and further investigation:

- **Domain-Specific Neural Models:** Future work could investigate transformer rerankers or embedding models fine-tuned specifically on scientific and aeronautical corpora rather than relying on zero-shot pretrained models trained on general web data.
- **Hybrid Lexical-Semantic Retrieval:** More advanced hybrid systems combining strong lexical retrieval (such as BM25+) with lightweight semantic expansion or dense retrieval could potentially improve recall while preserving lexical precision.
- **Contextual Query Expansion:** The current LSA and WordNet-based query expansion methods could be extended using contextual embeddings such as BERT, SciBERT, or Sentence Transformers to generate semantically richer and domain-aware expansions.
- **Larger Technical Corpora:** Many semantic and neural retrieval methods require substantially larger datasets to realize their full potential. Evaluating these approaches on larger scientific retrieval benchmarks may yield different conclusions.
- **Pseudo-Relevance Feedback (PRF):** Future systems could incorporate automatic relevance feedback mechanisms such as Rocchio-based expansion or probabilistic pseudo-relevance feedback to iteratively refine query representations.
- **Dense and Sparse Hybrid Retrieval:** Recent retrieval architectures combining sparse lexical retrieval with dense vector search could be explored to balance exact terminology matching with semantic generalization.
- **Improved Learning-to-Rank Features:** Additional ranking signals such as semantic similarity scores, positional features, citation information, or

contextual embeddings may improve the effectiveness of supervised ranking systems.

– **Cross-Domain Evaluation:** Future studies could compare the behavior of these retrieval systems across technical, biomedical, legal, and general-domain datasets to better understand how corpus characteristics influence retrieval performance.